

1. Dimension of $\frac{1}{\mu_0 \epsilon_0}$ should be equal to

[2023 (08 Apr Shift 1)]

- (1) $L T^{-1}$
- (2) $T^2 L^{-2}$
- (3) $L^2 T^{-2}$
- (4) $T L^{-1}$

2. Match List I with List II

	List I		List II
A.	Torque	I.	$M L^{-2} T^{-2}$
B.	Stress	II.	$M L^2 T^{-2}$
C.	Pressure gradient	III.	$M L^{-1} T^{-1}$
D.	Coefficient of viscosity	IV.	$M L^{-1} T^{-2}$

Choose the correct answer from the options given below :

[2023 (08 Apr Shift 2)]

- (1) A-II, B-I, C-IV, D-III
- (2) A-IV, B-II, C-III, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-IV, C-I, D-II

3. Given below are two statements :

Statement I : Astronomical unit (Au), Parsec (Pc) and Light year (ly) are units for measuring astronomical distances.

Statement II : $Au < Parsec (Pc) < ly$

In the light of the above statements, choose the most appropriate answer from the options given below:

[2023 (11 Apr Shift 1)]

- (1) Both Statements I and Statements II are incorrect
- (2) Statements I is correct but Statements II is incorrect
- (3) Both Statements I and Statements II are correct
- (4) Statements I is incorrect but Statements II is correct

4. If force (F), velocity (V) and time (T) are considered as fundamental physical quantity, then dimensional formula of density will be :

[2023 (11 Apr Shift 2)]

- (1) $F V^4 T^{-6}$
- (2) $F V^{-4} T^{-2}$
- (3) $F^2 V^{-2} T^6$
- (4) $F V^{-2} T^2$

5. Match List I with List II

	List-I		List-II
A	Spring constant	I	$[T^{-1}]$
B	Angular speed	II	$[MT^{-2}]$
C	Angular momentum	III	$[ML^2]$
D	Moment of Inertia	IV	$[ML^2T^{-1}]$

Choose the correct answer from the options given below:

[2023 (12 Apr Shift 1)]

- (1) A-I, B-III, C-II, D-IV
- (2) A-IV, B-I, C-III, D-II
- (3) A-II, B-I, C-IV, D-III
- (4) A-II, B-III, C-I, D-IV

6. In the equation $\left[X + \frac{a}{Y^2}\right][Y - b] = RT$, X is pressure, Y is volume, R is universal gas constant and T is temperature. The physical quantity equivalent to the ratio $\frac{a}{b}$ is:

[2023 (13 Apr Shift 2)]

- (1) Pressure gradient
- (2) Energy
- (3) Impulse
- (4) Coefficient of viscosity

7. The speed of a wave produced in water is given by $v = \lambda^a g^b \rho^c$. Where λ , g and ρ are wavelength of wave, acceleration due to gravity and density of water respectively. The values of a , b and c respectively, are

[2023 (15 Apr Shift 1)]

- (1) 1, -1, 0
- (2) $\frac{1}{2}$, 0, $\frac{1}{2}$
- (3) 1, 1, 0
- (4) $\frac{1}{2}$, $\frac{1}{2}$, 0

ANSWER KEYS

1. (3) 2. (3) 3. (2) 4. (2) 5. (3) 6. (2) 7. (4)

1. (3)

The speed of light can be written as

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \dots (1)$$

Rearrange equation (1) to obtain the required quantity.

$$\frac{1}{\mu_0 \epsilon_0} = c^2 \dots (2)$$

Use the method of dimensional analysis to obtain the dimension of the required quantity.

$$\left[\frac{1}{\mu_0 \epsilon_0} \right] = [LT^{-1}]^2 \\ = L^2 T^{-2}$$

2. (3)

The dimensions of the quantities are as follows.

$$F = M L T^{-2}$$

$$r = L$$

$$v = L T^{-1}$$

$$A = L^2$$

The dimensional form of stress is same as pressure.

$$\text{Torque} = \text{Force} \times \text{distance} = M L^2 T^{-2}$$

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} = M L^{-1} T^{-2}$$

$$\text{Pressure Gradient} = \frac{dp}{dx} = M L^{-2} T^{-2}$$

$$\eta = \frac{F}{6\pi r v} = \frac{M L T^{-2}}{L L T^{-1}} = M L^{-1} T^{-1}$$

3. (2)

Astronomical unit denotes roughly the distance from Earth to the Sun. Parsec is a unit of length used to measure the large distances to astronomical objects outside the Solar System. A light-year is another large unit of length used to express astronomical distances.

The relation between these units are as follows:

$$1 \text{ Parsec} = 2 \times 10^5 \text{ Au}$$

$$1 \text{ Au} = 1.58 \times 10^{-5} \text{ ly}$$

$$\text{Thus, } 1 \text{ Au} < 1 \text{ ly} < \text{Parsec}.$$

Hence, Statements I is true but Statements II is false.

4. (2)

The dimensional formula of force is

$$F = M L T^{-2}$$

The dimensional formula of velocity is

$$V = L T^{-1}$$

The formula for density is

$$\text{Density} = \frac{M}{L^3} = M L^{-3}$$

Comparing the density dimensions with the assumed fundamental quantities in the question,

$$M L^{-3} = [F]^x [V]^y [T]^z \\ \Rightarrow M L^{-3} = [M L T^{-2}]^x [L T^{-1}]^y [T]^z$$

$$\Rightarrow M L^{-3} = M^x L^{(x+y)} T^{-2x-y+z}$$

Comparing the exponents,

$$x = 1 \dots (i)$$

$$x + y = -3 \dots (ii)$$

$$-2x - y + z = 0 \dots (iii)$$

Solving equations (i), (ii) & (iii), we get

$$y = -4, z = -2.$$

$$\text{The dimensions of density} = F V^{-4} T^{-2}.$$

5. (3)

The formula to calculate the spring constant is

$$F = kx \dots (1)$$

The formula to calculate the angular speed is

$$\omega = \frac{2\pi}{T} \dots (2)$$

The formula to calculate the angular momentum is

$$L = mvr \dots (3)$$

The formula to calculate the moment of inertia is

$$I = MR^2 \dots (4)$$

Use the method of dimensional analysis to obtain the dimensions of the required quantities. From above relations, we can write

$$[MLT^{-2}] = [k][L]$$

$$\Rightarrow [k] = [MT^{-2}]$$

$$[\omega] = \frac{1}{[T]} \\ = [T^{-1}]$$

$$[L] = [M][LT^{-1}][L] \\ = [ML^2T^{-1}]$$

$$[I] = [M][L^2] \\ = [ML^2]$$

6. (2)

X & $\frac{a}{Y^2}$ have the same dimensions. Y & b also have the same dimensions.

Therefore,

$$\frac{a}{Y^2} = ML^{-1}T^{-2}$$

$$\Rightarrow a = ML^{-1}T^{-2} \times L^6 = ML^5T^{-2}$$

and $b = L^3$.

So, the dimensions of $\frac{a}{b}$ is,

$$\frac{a}{b} = \frac{ML^5T^{-2}}{L^3} = ML^2T^{-2}.$$

This is the dimensional form of energy. Hence, this is the right option.

7. (4)

The dimensional form of velocity can be written as

$$[v] = [\lambda^a g^b \rho^c]$$

$$\Rightarrow [M^0LT^{-1}] = [L]^a [LT^{-2}]^b [ML^{-3}]^c$$

Comparing the exponents for M, L, T ,

For M ,

$$M^c = M^0$$

For L ,

$$L^{a+b-3c} = L^1$$

For T ,

$$T^{-2b} = T^{-1}$$

So,

$$c = 0 \dots (i)$$

$$a + b - 3c = 1 \dots (ii)$$

$$-2b = -1 \dots (iii)$$

From equations (i),(ii),(iii),

$$a = \frac{1}{2}, b = \frac{1}{2}, c = 0$$